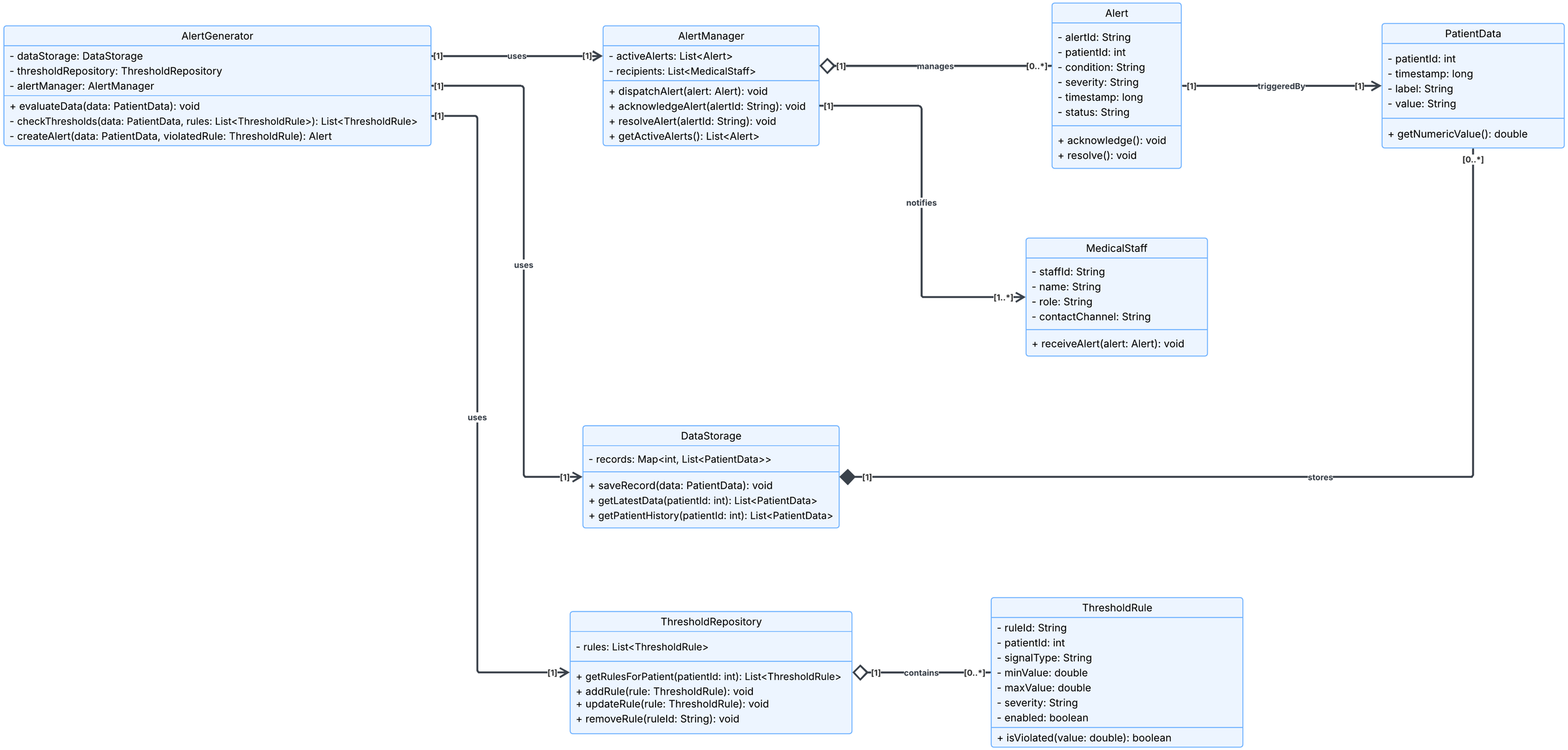
CHMS Alert Generation System – Class Diagram



This diagram models how the CHMS detects abnormal patient vitals and routes alerts to medical staff. The design is modular so that data evaluation, threshold management, alert creation, and alert dispatch are handled by separate classes.

AlertGenerator is the main evaluation component. It receives PatientData, retrieves relevant rules from ThresholdRepository, checks whether any threshold is violated, and creates an Alert when needed. This keeps the alert logic centralized and easy to modify. ThresholdRule represents personalized conditions for each patient, such as minimum or maximum acceptable values for a signal. ThresholdRepository stores these rules separately, which improves maintainability and allows future extension.

AlertManager handles dispatching alerts and tracking active ones. This avoids mixing evaluation with communication logic. MedicalStaff is modeled as the receiver of alerts, reflecting the real hospital workflow. Alert stores the key details of an alert, including patient ID, condition, severity, timestamp, and status, which supports traceability.

The subsystem also connects to PatientData and DataStorage. PatientData represents the incoming measurement that may trigger an alert, while DataStorage provides access to stored data when needed. Overall, the diagram emphasizes separation of concerns, personalized monitoring, and clear alert routing.